

Notes: Ferraz and Finan (AER, 2011)

Electoral Accountability and Corruption: Evidence from the Audits of Local Governments

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1 Big picture

- **Question.** Does electoral accountability reduce political corruption? The paper asks whether mayors who can run for reelection behave less corruptly than mayors who face a binding term limit.
- **Setting.** The setting is Brazilian municipal government. A constitutional amendment allowed mayors to serve two consecutive terms, and a federal anti-corruption program randomly audited municipalities for misuse of federal funds.
- **Data.** The authors use detailed audit reports from the Controladoria Geral da Uniao (CGU) for 476 municipalities audited before the 2004 municipal elections. They manually code corruption from the audit narratives.
- **Main outcome.** The preferred corruption measure is the share of audited federal resources associated with fraud in public procurement, diversion of funds, or overinvoicing.
- **Core comparison.** First-term mayors can run again and therefore face reelection incentives; second-term mayors face a term limit and are effectively lame ducks.
- **Main result.** First-term mayors misappropriate substantially fewer resources. The preferred estimate implies that reelection incentives reduce the share of stolen audited resources by about 2.7 percentage points, roughly 27 percent relative to second-term mayors.
- **Identification strategy.** The baseline OLS comparison controls for mayor, municipal, political, state, and audit-lottery characteristics. The paper then strengthens the design using close-election regression discontinuity, ability checks, and experience checks.
- **Mechanism.** The effect is stronger where local accountability is otherwise weaker: in places without local media, without local judicial presence, and where political competition is stronger.
- **Economic takeaway.** Reelection incentives are not only formal electoral rules; they are an enforcement technology. When voters can potentially punish bad behavior, politicians reduce visible rent extraction.

2 Incremental contribution of the paper

- **From perceived corruption to measured corruption.** Much earlier work uses cross-country corruption perception indexes. This paper constructs direct, municipality-level measures from audit reports.
- **From cross-country institutions to within-country identification.** The paper keeps national institutions fixed and uses variation across Brazilian mayors' electoral horizons.
- **A political-agency test with objective outcomes.** The paper gives a direct empirical test of political agency models in which elections discipline incumbents.

- **A term-limit design.** The 1997 constitutional reform creates a distinction between first-term mayors with reelection incentives and second-term mayors facing term limits.
- **Mechanism evidence.** The paper does not stop at the average effect. It examines whether effects vary with media, judiciary presence, political competition, and legislative support.
- **Alternative explanation checks.** The paper explicitly addresses selection, political ability, political experience, auditor manipulation, and the possibility that mayors simply learn corruption over time.
- **Broader contribution.** The paper shows that accountability institutions can reduce corruption even in environments where corruption is common and local elites are politically influential.

3 Institutional background

3.1 Brazilian municipalities and mayoral incentives

Brazilian municipalities receive large transfers from the federal government and are responsible for important public services such as education, health, transport, and infrastructure. Mayors play a central role in allocating these resources, proposing budgets, and executing approved spending.

A 1997 constitutional amendment allowed mayors to run for a second consecutive term. This creates a clean distinction during the 2001–2004 electoral term: first-term mayors can seek reelection, while second-term mayors face a binding term limit.

Interpretation: The institutional setting maps naturally into a political agency framework. A first-term mayor may restrain rent extraction to improve reelection chances; a second-term mayor has weaker direct electoral discipline.

3.2 Brazil’s anti-corruption audit program

In 2003 the CGU began randomly auditing municipalities for the use of federal funds. The audits inspect municipal accounts, procurement documents, public works, and service delivery. The reports are sent to oversight institutions and summarized publicly.

Interpretation: Random audits create unusually credible data on local corruption. The paper’s empirical strength comes from combining this audit evidence with variation in mayors’ electoral horizons.

4 Theory and empirical predictions

4.1 Political agency logic

The paper uses a simple two-period political agency model. There are corrupt and non-corrupt politicians. Noncorrupt politicians choose policies aligned with voters. Corrupt

politicians receive private rents from choosing actions that deviate from voters' interests. In period 2, there are no future reelection incentives, so corrupt politicians extract rents. In period 1, a corrupt politician faces a trade-off: extract rents now or behave like a noncorrupt politician to increase reelection chances and obtain future office benefits.

A simplified rent-extraction comparison is:

$$E[r_2 \mid \text{second term}] > E[r_1 \mid \text{first term with reelection incentives}],$$

provided that the disciplining value of reelection is sufficiently large.

Main prediction: first-term mayors should commit less visible corruption than second-term mayors.

4.2 Additional predictions

- **Information.** If voters have more information about politicians, the selection and discipline effects change. Empirically, the difference between first- and second-term mayors should vary with local media.
- **Judicial punishment.** If corruption is more likely to be prosecuted, expected rents from corruption fall. The difference should vary with local judicial institutions.
- **Political competition.** If a mayor has a safe electoral advantage, reelection incentives discipline him less. Accountability should be stronger in more competitive environments.
- **Visibility.** Reelection incentives should reduce visible corruption more than nonvisible mismanagement or administrative violations that voters cannot observe or punish.

5 Data and measurement

5.1 Audit-based corruption measures

The authors code each audit report and classify political corruption into three main categories:

1. **Diversion of funds:** federal resources disappear or goods/services are claimed but not delivered.
2. **Illegal procurement:** bidding processes are manipulated, often involving phantom firms or politically connected suppliers.
3. **Overinvoicing:** goods or services are purchased above market value.

The preferred dependent variable is:

$$\text{Corruption share}_i = \frac{\text{audited resources associated with corruption}_i}{\text{total audited resources}_i}.$$

The paper also uses alternative measures: the number of corruption irregularities and the share of audited service items involving corruption.

5.2 Mismanagement measure

The paper separately codes less visible administrative violations and potential passive waste as mismanagement. These include violations such as noncompetitive procurement where the public good is still provided, misuse of funds across categories, or failure to follow administrative rules.

Interpretation: This measure is important because it helps distinguish electoral accountability from pure experience. If second-term mayors merely learn how to abuse office, they should commit more of both visible corruption and nonvisible mismanagement. The data show otherwise.

5.3 Mayor and municipal controls

The empirical models include mayor characteristics, municipal socioeconomic characteristics, political institutions, audit-lottery fixed effects, and state fixed effects. These controls absorb many observable differences across first- and second-term mayor municipalities.

6 Empirical specifications

6.1 Baseline OLS design

The baseline regression is:

$$r_i = \beta I_i + X_i' \phi + Z_i' \gamma + \varepsilon_i,$$

where:

- r_i is the corruption level in municipality i ;
- I_i equals one if the mayor is in the first term;
- X_i includes municipal controls;
- Z_i includes mayor controls.

The coefficient β captures the difference in corruption between first- and second-term mayors after conditioning on observables. A negative β is consistent with reelection incentives reducing corruption.

6.2 Regression discontinuity around close elections

To address unobserved municipal-level confounds, the paper compares municipalities where incumbents barely won reelection in 2000 with municipalities where incumbents barely lost. The specification is:

$$r_i = \beta I_i + f(W_i) + X_i' \phi + Z_i' \gamma + \varepsilon_i,$$

with

$$I_i = 1[W_i \leq 0],$$

where W_i is the incumbent’s margin of victory. Close elections are used as quasi-random variation in whether the 2001–2004 mayor is first-term or second-term.

6.3 Ability and experience checks

The paper addresses two further concerns:

- **Political ability:** second-term mayors may differ because they won reelection. The paper compares them to first-term mayors who are later reelected or predicted to be reelected.
- **Political experience:** second-term mayors may know more about how to be corrupt. The paper compares them to first-term mayors with prior mayoral or legislative experience.

6.4 Matching grants specification

To test whether reelection incentives also affect positive political effort, the paper estimates:

$$y_{it} = \alpha_0 + \sum_t \alpha_t (T_t \times I_i) + \lambda_t + \eta_i + \epsilon_{it},$$

where y_{it} measures whether a municipality received a matching grant, the grant amount per capita, or the share of the grant received.

7 Identification and empirical design

7.1 What the design identifies

The paper estimates whether corruption is lower when mayors face direct reelection incentives. The cleanest interpretation is the effect of moving from a binding term-limit environment to a first-term environment in which reelection is possible.

7.2 Key threats

- **Municipal selection.** More accountable or richer municipalities may both reelect mayors differently and have less corruption. The close-election RDD addresses this concern.
- **Political ability.** Second-term mayors may be selected winners. The paper compares them to successful first-term mayors.
- **Experience and learning.** Second-term mayors may have learned how to steal. The paper compares them to politically experienced first-term mayors and tests visible versus nonvisible violations.
- **Auditor manipulation.** First-term mayors may try harder to influence audit reports. The paper tests whether estimates vary by election-year audits or political alignment with federal/state governments.

7.3 Interpretation

The design is strongest as evidence that electoral horizons discipline visible rent extraction. The evidence is less suited to identifying the globally optimal term-limit rule, because the paper observes a two-term regime relative to a one-term regime but cannot observe what would happen under unlimited reelection.

8 Tables and figures

8.1 Figure 1: Electoral cycles and audit timing

Read:

- Figure 1 summarizes the institutional timing: municipal elections in 1996, 2000, and 2004; the 1997 reform allowing reelection; and the CGU audit program introduced during the 2001–2004 term.
- The figure clarifies why the paper can compare first- and second-term mayors during the same audit window.

Economic message: The empirical design is built from an institutional overlap: reelection incentives vary across mayors while the audit program measures corruption during the same governing period.

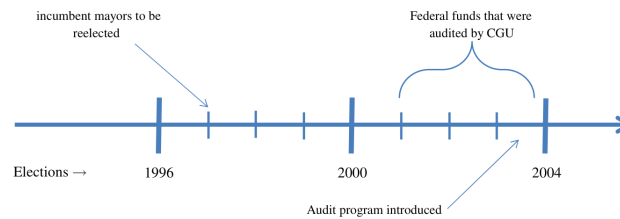


FIGURE 1. TIMELINE OF THE ELECTORAL CYCLES AND AUDIT TIMING

Notes: Figure shows the timing of Brazil's municipal elections and the introduction of the audit program. It also depicts the period over which the federal funds were audited and when the constitutional amendment allowing for a second consecutive term in office took place.

Figure 1: Timeline of electoral cycles and audit timing.

8.2 Tables 1 and 2: Corruption measurement and first-versus-second term differences

Read:

- Table 1 reports that 79 percent of municipalities have at least one corruption irregularity; diversion of funds and illegal procurement are the most common categories.
- Conditional on corruption, the average value of irregularities is large relative to audited resources.
- Table 2 shows raw differences: second-term mayors have a higher share of audited resources associated with corruption, especially through illegal procurement.

Economic message: The raw data already point to the central pattern: lame-duck mayors appear more corrupt. The rest of the paper tests whether this pattern reflects reelection incentives rather than selection, ability, or experience.

TABLE 1—SUMMARY STATISTICS OF CORRUPTION BY TYPE

	Type of irregularity				
	Diversion of funds	Illegal procurement	Overinvoicing	Corruption indicator	Mismanagement
<i>Proportion of municipalities with at least one irregularity</i>	0.536 (0.499)	0.576 (0.495)	0.071 (0.258)	0.786 (0.411)	0.986 (0.116)
<i>Average number of irregularities</i>	1.686 (1.006)	1.657 (0.945)	1.029 (0.171)	2.457 (1.554)	n/a
<i>Average value of irregularity (R\$)</i>	159,205.20 (324,303.8)	291,431.50 (578,272.1)	60,670.14 (166,733.8)	327,573.10 (627,514.2)	n/a
<i>Share of audited resources</i>	0.041 (0.072)	0.070 (0.093)	0.015 (0.036)	0.080 (0.109)	n/a
<i>Share of audited items</i>	0.047 (0.036)	0.045 (0.028)	0.029 (0.012)	0.067 (0.050)	1.647 (1.154)

Notes: This table reports descriptive statistics on the various measures of corruption computed from the audit reports. These statistics presented in columns 1–4 were computed for our sample of 476 municipalities. In column 5, the statistics were computed for 366 municipalities. Standard deviations are reported in parentheses.

B. Summary Statistics on Corruption and Mismanagement

(a) Table 1: Summary statistics of corruption by type

TABLE 2—SUMMARY STATISTICS OF CORRUPTION BY FIRST- AND SECOND-TERM MAYORS

	Share of audited resources			Incidence of irregularities			Share of audited items		
	First term (1)	Second term (2)	Difference (3)	First term (4)	Second term (5)	Difference (6)	First term (7)	Second term (8)	Difference (9)
Diversion of funds	0.021	0.022	0.001 [0.005]	0.852	0.971	0.119 [0.103]	0.024	0.026	0.002 [0.003]
Illegal procurement	0.033	0.050	0.017 [0.007]	0.837	1.107	0.270 [0.100]	0.023	0.029	0.006 [0.003]
Overinvoicing	0.001	0.002	0.001 [0.001]	0.074	0.073	−0.001 [0.025]	0.002	0.002	0.000 [0.001]
Corruption	0.055	0.074	0.019 [0.009]	1.763	2.150	0.388 [0.157]	0.050	0.057	0.008 [0.005]

Notes: This table compares the various measures of corruption between first- and second-term mayors. These statistics were computed only for the 476 municipalities. Columns 1, 4, and 7 report the means for the 270 municipalities with a first-term mayor. Columns 2, 5, and 8 report the mean for the 206 municipalities with a second-term mayor. Columns 3, 6, and 9 report the difference in means, and robust standard errors of the difference are presented in brackets.

commit 0.12 and 0.27 more irregularities in the diversion of funds and illegal procurement practices, respectively, which represents 0.2 and 0.6 percentage point differences in the share of services items audited.

C. Municipal and Mayor Characteristics

(b) Table 2: Corruption by first- and second-term mayors

8.3 Table 3: Balance on mayor and municipal characteristics

Read:

- Table 3 compares first- and second-term mayor municipalities on mayor characteristics, socioeconomic conditions, and political variables.
- Most mayor and municipal characteristics are similar across the two groups.
- The main differences appear in political variables such as legislative support and margin of victory, motivating the inclusion of political controls and close-election analysis.

Economic message: The first- and second-term municipalities are not experimentally assigned, but the observable balance is reasonably strong, and the remaining differences are addressed through controls and robustness designs.

TABLE 3— SUMMARY STATISTICS OF MAYORS AND MUNICIPALITIES

	First-term mayors	Second-term mayors	Difference
<i>Mayor characteristics</i>			
Male	0.95	0.96	-0.01 [0.020]
Schooling	6.10	6.07	0.03 [0.176]
Age	47.86	46.67	1.19 [0.848]
<i>Municipal characteristics</i>			
Population	25,828.74	24,878.19	950.54 [3,877.32]
% urban population	0.62	0.58	0.046 [0.021]**
% adults with secondary school	0.25	0.22	0.028 [0.010]***
New municipality	0.21	0.27	-0.055 [0.039]
GDP per capita (R\$1,000)	5,309.95	6,189.44	-879.490 [1,226.10]
Gini coefficient	0.57	0.57	0.007 [0.005]
Intergovernmental transfers (R\$ million)	12.50	11.90	0.56 [1.943]
Participatory budgeting	0.03	0.03	0.008 [0.028]
<i>Political characteristics</i>			
% legislators in mayor's party	0.27	0.36	-0.087 [0.014]***
Legislators per voter (×100)	0.14	0.15	-0.01 [0.01]
Effective number parties legislature	4.55	3.91	0.643 [0.145]***
Margin of victory in 2000 elections	0.15	0.23	-0.083 [0.018]***
Judiciary district	0.57	0.52	0.046 [0.046]
Media	0.84	0.82	0.017 [0.035]
Number of audited items	41.50	42.72	-1.221 [1.894]
Total resources audited (R\$ million)	5.86	5.18	0.68 [0.75]

Notes: This table presents a comparison of the mean political, mayor, and socioeconomic characteristics of the municipalities between first- and second-term mayors. These statistics were computed only for the 476 municipalities that contained the full set of nonmissing characteristics. Column 1 reports the means for the 270 municipalities with a first-term mayor. Column 2 reports the mean for the 206 municipalities with second-term mayors. Column 3 reports the difference in means, and robust standard errors of the difference are presented in brackets.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

Table 3: Summary statistics of mayors and municipalities.

8.4 Tables 4 and 5: Main corruption estimates and alternative corruption measures

Read:

- Table 4 is the main result. Across specifications, first-term mayors are associated with a lower share of audited resources involving corruption.
- The preferred OLS estimate is about -0.027 with state and audit-lottery intercepts and a rich set of controls.
- Table 5 shows robustness to alternative outcomes: number of corruption irregularities and share of audited items involving corruption.

Economic message: The main estimate is not a measurement artifact. Reelection incentives reduce corruption across multiple corruption definitions and estimation methods.

Dependent variable	Share of audited resources involving corruption							
	OLS (1)	OLS (2)	OLS (3)	OLS (4)	OLS (5)	OLS (6)	Matching (7)	Tobit (8)
Mayor in first term	-0.019 [0.009]**	-0.020 [0.010]**	-0.020 [0.010]**	-0.024 [0.011]**	-0.026 [0.011]**	-0.027 [0.011]**	-0.028 [0.010]**	-0.042 [0.012]**
R^2	0.01	0.08	0.10	0.12	0.14	0.20	n/a	n/a
Observations	476	476	476	476	476	476	476	476
Mayor characteristics	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Municipal characteristics	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Political and judicial institutions	No	No	No	Yes	Yes	Yes	Yes	Yes
Lottery intercepts	No	No	No	No	Yes	Yes	Yes	Yes
State intercepts	No	No	No	No	No	Yes	Yes	Yes

Notes: This table reports the effects of reelection incentives on share of resources found to involve corruption. Columns 1–6 present the results of an OLS regression of the dependent variables listed in that column on an indicator variable for whether the mayor is in his first term. Column 7 estimates the relationship using Abadie and Imbens (2004) matching estimator, and column 8 uses a Tobit specification and reports marginal effects. The log likelihood for the Tobit estimation is 194.20. Mayor characteristics include the age, gender, education, and party affiliation. Municipal characteristics: population expressed in logarithms, percentage of the population that has at least a secondary education, percentage of the population that lives in the urban sector, new municipality, log GDP per capita per in 2002, Gini coefficient. Political and judicial institutions include: effective number of political parties in the legislature, the number of legislators divided by the number of voters, the share of the legislature that is of the same party as the mayor, and whether the municipality is judiciary district, and the amount of resources sent to the municipality expressed in logarithms. Robust standard errors are displayed in brackets.

***Significant at the 1 percent level.
 **Significant at the 5 percent level.
 *Significant at the 10 percent level.

A. Main Results on Corruption

(a) Table 4: Main effects on corruption share

(b) Table 5: Alternative corruption measures

Dependent variable	Numbers of irregularities involving corruption				Share of audited items involving corruption			
	OLS (1)	OLS (2)	Matching (3)	Negative binomial (4)	OLS (5)	OLS (6)	Matching (7)	Tobit (8)
Mayor in first term	-0.388 [0.158]**	-0.471 [0.148]***	-0.339 [0.146]**	-0.456 [0.127]**	-0.008 [0.005]	-0.011 [0.004]**	-0.010 [0.005]*	-0.009 [0.005]*
R^2	0.01	0.43	n/a	n/a	0.01	0.45	n/a	n/a
Observations	476	476	476	476	476	476	476	476
Mayor characteristics	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Municipal characteristics	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Political and judicial institutions	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Lottery intercepts	No	Yes	Yes	Yes	No	Yes	Yes	Yes
State intercepts	No	Yes	Yes	Yes	No	Yes	Yes	Yes

Notes: This table reports the effects of reelection incentives on number of irregularities associated with corruption and the share of service items involving corruption. Columns 1–2 and columns 5–6 present the results of an OLS regression of the dependent variables listed in that column on an indicator variable for whether the mayor is in his first term. Columns 3 and 7 estimate the relationship using Abadie and Imbens (2004) matching estimator; column 4 uses a negative binomial specification and column 8 uses a Tobit specification. In both columns 4 and 8, marginal effects are reported. The pseudo-likelihood for the negative binomial specification is -729.68 and the log likelihood for the Tobit is 459.40. Mayor characteristics include the age, gender, education, and party affiliation. Municipal characteristics: population expressed in logarithms, percentage of the population that has at least a secondary education, percentage of the population that lives in the urban sector, new municipality, log GDP per capita per in 2002, Gini coefficient. Political and judicial institutions include: effective number of political parties in the legislature, the number of legislators divided by the number of voters, the share of the legislature that is of the same party as the mayor, whether the municipality is judiciary district, and the amount of resources sent to the municipality expressed in logarithms. Robust standard errors are displayed in brackets.

***Significant at the 1 percent level.
 **Significant at the 5 percent level.
 *Significant at the 10 percent level.

8.5 Table 6 and Figure 2: Close-election design

Read:

- Table 6 restricts attention to incumbents who ran for reelection and estimates models around the incumbent win/loss cutoff.
- Across linear, quadratic, cubic, and spline specifications, first-term mayors remain less corrupt.
- Figure 2 visualizes a discontinuous increase in corruption at the threshold where the incumbent barely wins and becomes a second-term mayor.

Economic message: The RD evidence reduces concerns that unobserved municipal characteristics drive the results. Around close elections, the assignment of first- versus second-term mayor is plausibly quasi-random.

TABLE 6—THE EFFECT OF REELECTION INCENTIVES ON CORRUPTION, CONTROLLING FOR UNOBSERVED MUNICIPAL CHARACTERISTICS

Dependent variable	Share of audited resources involving corruption						
	Incumbents who ran for reelection in 2000 (1)	Linear (2)	Quadratic (3)	Cubic (4)	Linear spline (5)	Quadratic spline (6)	Cubic spline (7)
Mayor in first term	-0.031 [0.014]**	-0.039 [0.019]**	-0.040 [0.019]**	-0.038 [0.022]*	-0.043 [0.019]**	-0.047 [0.024]**	-0.028 [0.029]
Observations	328	328	328	328	328	328	328
R^2	0.27	0.27	0.27	0.27	0.27	0.28	0.28
Mayor characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Municipal characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Political and judicial institutions	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Lottery intercepts	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State intercepts	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table reports the effects of reelection incentives on the share of resources involving corruption. The estimation samples in columns 1–7 include only municipalities where the incumbent ran for reelection. In columns 2–4, the functional form assumption for the running variable is specified at the top of each column. Mayor characteristics include the age, gender, education, and party affiliation of the mayor. Municipal characteristics: population expressed in logarithms, percentage of the population that has at least a secondary education, percentage of the population that lives in the urban sector, new municipality, log GDP per capita per in 2002, Gini coefficient. Political and judicial institutions include: effective number of political parties in the legislature, the number of legislators divided by the number of voters, the share of the legislature that is of the same party as the mayor, and whether the municipality is judiciary district, and the amount of resources sent to the municipality expressed in logarithms. Robust standard errors are displayed in brackets.

***Significant at the 1 percent level.
 **Significant at the 5 percent level.
 *Significant at the 10 percent level.

(a) Table 6: RD specifications around close elections

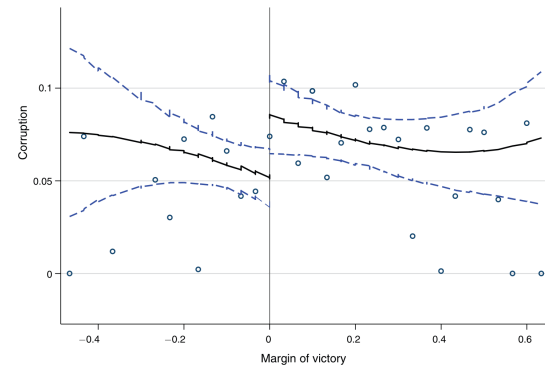


FIGURE 2. THE EFFECTS OF REELECTION INCENTIVES ON CORRUPTION

Notes: Figure shows the share of audited resources involving corruption by the margin of victory for incumbents who ran for reelection in 2000. Each figure presents mean corruption for a bin size of 30 percentage points (hollow circles) along with the fitted values from a third degree polynomial fit on each side of the discontinuity. The dashed lines denote 90 percent confidence intervals. These estimates were computed for a sample size of 328 municipalities.

(b) Figure 2: Discontinuity in corruption at the reelection threshold

8.6 Table 7: Ability and experience checks

Read:

- Table 7 compares second-term mayors to first-term mayors who are later reelected or predicted to be reelected, addressing political ability.
- It also controls for prior political experience and compares second-term mayors to experienced first-term mayors.
- The first-term coefficient remains negative and similar in magnitude.

Economic message: The results are unlikely to be explained by second-term mayors being more politically able or merely more experienced.

TABLE 7—THE EFFECT OF REELECTION INCENTIVES ON CORRUPTION CONTROLLING FOR ABILITY AND EXPERIENCE

Dependent variable	Share of audited resources involving corruption					
	Second-term and first-term later reelected (1)	Second-term and first-term later reelected predicted (2)	Full sample (3)	Full sample (4)	Second-term and first-term that served as previous mayors (5)	Second-term and first-term that served as mayor or legislator in past (6)
Mayor in first-term	−0.04 [0.013]***	−0.034 [0.018]*	−0.027 [0.012]**	−0.030 [0.012]**	−0.038 [0.014]***	−0.027 [0.017]
Mayor with political experience			−0.007 [0.011]			
Number of years in political office				0.008 [0.007]		
Number of years in political office ²				−0.002 [0.001]		
R ²	0.27	0.29	0.21	0.21	0.30	0.29
Observations	313	294	476	476	287	311
Mayor characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Municipal characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Political and judicial institutions	Yes	Yes	Yes	Yes	Yes	Yes
Lottery intercepts	Yes	Yes	Yes	Yes	Yes	Yes
State intercepts	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table reports the effects of reelection incentives on the share of resources involving corruption. Column 1 compares the corruption levels of second term mayors to those of the subset of first term mayors that were reelected in the subsequent mayor elections. Column 2 compares the corruption levels of second term mayors to those of the subset of first term mayors who were predicted to be reelected, based on a propensity score. Column 3–4 is estimated on the full sample. Column 5 includes only municipalities with a second-term mayor and first-term mayors who had a mayor in a previous term. Column 6 includes only municipalities with a second-term mayor and first-term mayors who had been either a mayor or legislator in a previous term. Mayor characteristics include the age, gender, education and party affiliation of the mayor. Municipal characteristics: population expressed in logarithms, percentage of the population that has at least a secondary education, percentage of the population that lives in the urban sector, new municipality, log GDP per capita per in 2002, Gini coefficient. Political and judicial institutions include: effective number of political parties in the legislature, the number of legislators divided by the number of voters, the share of the legislature that is of the same party as the mayor, and whether the municipality is judiciary district, and the amount of resources sent to the municipality expressed in logarithms. Robust standard errors are displayed in brackets.

***Significant at the 1 percent level.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

second-term mayors to the subset of first-term mayors who were predicted to be reelected. After controlling for the effects of the audits, the point estimate reduces to −0.034 (robust standard error = 0.017) and is still significant at 90 percent confidence (see column 2).

D. Accounting for Political Experience

Even if we account for differences in ability between first- and second-term mayors, politicians in power for a longer period of time may learn to be more corrupt

status, age, and party affiliation dummies; the municipality's log population, population with secondary school education, age of municipality, log GDP per capita, income equality, share of the legislative branch that supports the mayor, effective number of parties in 2000 election, an indicator for whether there is a judge in the municipality, and state fixed effects. The predicted indicator is equal to one if the propensity score was greater than or equal to 0.5. The estimation predicted 64 percent of the cases correctly

Table 7: Controlling for political ability and experience.

8.7 Tables 8 and 9: Nonvisible violations and positive political effort

Read:

- Table 8 shows that reelection incentives do not reduce less visible mismanagement violations.
- This supports the electoral accountability mechanism because voters can punish visible corruption more easily than hidden administrative irregularities.
- Table 9 shows that first-term mayors become more likely to obtain matching grants as elections approach.

Economic message: Electoral incentives shape both restraint and effort: mayors facing reelection reduce visible rent extraction and increase visible public-good procurement.

TABLE 8—REELECTION INCENTIVES AND OTHER FORMS OF BAD GOVERNANCE

Dependent variable	Share of audited items involving mismanagement			Corruption
	Full sample (1)	Second- and first-term mayors later reelected (2)	Second- and first-term mayors who served previously as mayors (3)	Full sample (4)
Mayor in first term	0.054 [0.102]	0.069 [0.151]	0.088 [0.180]	−0.036 [0.013]***
Observations	366	243	218	366
R ²	0.44	0.50	0.49	0.28
Mayor characteristics	Yes	Yes	Yes	Yes
Municipal characteristics	Yes	Yes	Yes	Yes
Political and judicial institutions	Yes	Yes	Yes	Yes
Lottery intercepts	Yes	Yes	Yes	Yes
State intercepts	Yes	Yes	Yes	Yes

Notes: This table reports the effects of reelection incentives on the share of audited items found to involve acts of mismanagement. Column 1 uses the full sample. Column 2 compares the mismanagement levels of second-term mayors to those of the subset of first-term mayors who were reelected in the subsequent mayoral elections. Column 3 includes only municipalities with a second-term mayor and first-term mayors who were mayors in a previous term. Column 4 reports the effects of reelection incentives on corruption for the sample used in estimating the results presented in column 1. Mayor characteristics include the age, gender, education, and party affiliation of the mayor. Municipal characteristics: population expressed in logarithms, percentage of the population that has at least a secondary education, percentage of the population that lives in the urban sector, new municipality, log GDP per capita in 2002, Gini coefficient. Political and judicial institutions include: effective number of political parties in the legislature, the number of legislators divided by the number of voters, the share of the legislature that is of the same party as the mayor, whether the municipality is judiciary district, and the amount of resources sent to the municipality expressed in logarithms. Robust standard errors are displayed in brackets.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

for higher office in 2006. Thus, assuming that politicians expect low probabilities of returning to office and discount future gains, it is reasonable to expect that, on average, second-term mayors will act as if they face real term limits.

Moreover, if mayors respond to reelection opportunities, they will not only refrain from rent extraction, but will also exert effort along other dimensions of political performance. In this subsection, we examine if reelection incentives affect whether mayors procure matching grants from the central government which is the only

TABLE 9—REELECTION INCENTIVES AND MATCHING GRANTS

Dependent variable	Received a grant		Amount of grants per capita		Share of the grant received	
	(1)	(2)	(3)	(4)	(5)	(6)
Mayor in first term × Election(t)	0.140 [0.049]***	0.132 [0.057]**	2.053 [0.594]***	2.002 [0.688]***	0.158 [0.050]***	0.143 [0.059]**
Mayor in first term × Election(t−1)	0.105 [0.047]**	0.100 [0.054]*	1.528 [0.572]**	1.494 [0.655]**	0.118 [0.048]**	0.109 [0.055]**
Mayor in first term × Election(t−2)	0.080 [0.035]**	0.078 [0.040]*	1.342 [0.425]***	1.325 [0.491]***	0.043 [0.036]	0.038 [0.042]
Mayor in first term	−0.058 [0.034]*		−1.107 [0.423]**		−0.061 [0.034]*	
F-test: Election(t−1) = Election(t−2)	0.38	0.23	0.15	0.09	2.91	1.95
F-test: Election(t) = Election(t−1)	0.59	0.38	0.95	0.66	0.82	0.44
F-test: Election(t) = Election(t−3)	2.15	1.33	2.00	1.36	7.44	4.47
Observations	1,904	1,904	1,904	1,904	1,903	1,903
Number of municipalities	476	476	476	476	476	476
R ²	0.13	0.45	0.14	0.48	0.12	0.44
Year intercepts	Y	Y	Y	Y	Y	Y
Municipal and mayor characteristics	Y	n/a	Y	n/a	Y	n/a
Municipality fixed-effects	N	Y	N	Y	N	Y
Municipal and mayor characteristic × time trends	N	Y	N	Y	N	Y

Notes: This table reports the effects of reelection incentives on the procurement of matching grants. The specification presented in the odd columns includes municipal and mayor characteristics and year intercepts, whereas the specifications in the even columns also include municipality fixed-effects and linear time trends interacted with our mayor and municipal covariates. The sample represents yearly data from 2001 to 2004 for each municipality in our sample. Robust standard errors clustered at the municipality level are displayed in brackets.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

(a) Table 8: Mismanagement versus corruption

(b) Table 9: Reelection incentives and matching grants

8.8 Tables 10 and 11: Local accountability conditions and audit manipulation checks

Read:

- Table 10 shows that local context matters. The difference between first- and second-term mayors is smaller where media or judiciary presence already disciplines politicians.
- The effect is stronger where political competition gives reelection incentives more bite.
- Table 11 tests whether the audit reports were politically manipulated. The evidence does not support the view that first-term mayors received more favorable reports

because of bribery or political alignment.

Economic message: The heterogeneity results strengthen the mechanism. Reelection incentives matter most when they are the relevant source of discipline, and the audit-manipulation checks reduce a major alternative interpretation.

TABLE 10—LOCAL CHARACTERISTICS AND THE EFFECT OF REELECTION INCENTIVES ON CORRUPTION

Dependent variable	Share of audited resources involving corruption			
	(1)	(2)	(3)	(4)
Mayor in first term	−0.049 [0.017]***	−0.098 [0.029]***	−0.066 [0.025]***	−0.045 [0.022]**
First term × judiciary district	0.037 [0.020]*			
Judiciary district	−0.029 [0.018]			
First term × media		0.084 [0.029]**		
Media		−0.051 [0.026]**		
First term × legislative support			0.120 [0.060]**	
Legislative support			−0.138 [0.053]***	
First term × political competition				0.079 [0.079]
Political competition				−0.149 [0.097]
F-test joint hypothesis	4.05	5.87	3.83	3.33
P-value	0.02	0.00	0.02	0.04
Observations	476	476	476	476
R ²	0.21	0.23	0.21	0.21
Mayor characteristics	Yes	Yes	Yes	Yes
Municipal characteristics	Yes	Yes	Yes	Yes
Political and judicial institutions	Yes	Yes	Yes	Yes
Lottery intercepts	Yes	Yes	Yes	Yes
State intercepts	Yes	Yes	Yes	Yes

Notes: This table reports the effects of reelection incentives on share of resources involving corruption. The sample includes all 476 municipalities. All regressions include mayor characteristics, municipal characteristics, political and judiciary characteristics, and state and lottery intercepts. Political competition is defined as 1 divided by the effective number of political parties. Media is defined as an indicator for whether the municipality has a local AM radio station or newspaper. Mayor characteristics include the age, gender, education, and party affiliation of the mayor. Municipal characteristics: population expressed in logarithms, percentage of the population that has at least a secondary education, percentage of the population that lives in the urban sector, new municipality, log GDP per capita 2002, Gini coefficient. Political and judicial institutions include: effective number of political parties in the legislature, the number of legislators divided by the number of voters, the share of the legislature that is of the same party as the mayor, whether the municipality is judiciary district, and the amount of resources sent to the municipality expressed in logarithms. Robust standard errors are displayed in brackets.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

TABLE 11—TESTING FOR THE POLITICAL MANIPULATION OF AUDIT REPORTS

Dependent variable	Share of audited resources indicated as corruption			Log(value projects audited)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Mayor in first term	−0.03 [0.014]**	−0.029 [0.012]**	−0.032 [0.014]**	0.055 [0.122]	0.091 [0.168]	0.049 [0.123]	0.100 [0.164]
First term × audited in election year	0.003 [0.017]				−0.093 [0.176]		
First term × mayor in PT		0.022 [0.050]				0.338 [0.357]	
First term × same party governor			0.012 [0.020]				−0.153 [0.206]
Mayor in PT	−0.026 [0.021]	−0.043 [0.048]	−0.027 [0.022]	−0.363 [0.191]*	−0.365 [0.190]*	−0.621 [0.342]*	−0.348 [0.197]*
Mayor same party governor			−0.01 [0.017]				0.102 [0.168]
Observations	476	476	476	476	476	476	476
R ²	0.20	0.21	0.21	0.61	0.61	0.61	0.61
Mayor characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Municipal characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Political and judicial institutions	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Lottery intercepts	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State intercepts	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The sample includes all 476 municipalities. All regressions include mayor characteristics, municipal characteristics, political and judiciary characteristics, state and lottery intercepts. Mayor characteristics include the age, gender, education, and party affiliation of the mayor. Municipal characteristics: population expressed in logarithms, percentage of the population that has at least a secondary education, percentage of the population that lives in the urban sector, new municipality, log GDP per capita 2002, Gini coefficient. Political and judicial institutions include: effective number of political parties in the legislature, the number of legislators divided by the number of voters, the share of the legislature that is of the same party as the mayor, whether the municipality is a judiciary district, and the amount of resources sent to the municipality expressed in logarithms. Robust standard errors are displayed in brackets.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

caught is high. According to program officials, there has never been an incidence in which auditors have even been caught receiving bribes.

(a) Table 10: Local characteristics and accountability

(b) Table 11: Political manipulation of audit reports

9 Short summary

This note summarizes the paper, which uses Brazilian CGU audit reports, municipal federal-transfer data, mayoral term-limit variation, first- versus second-term comparisons, close-election regression discontinuity, ability and experience checks, and heterogeneity by media, judiciary, and political competition to study whether electoral accountability reduces political corruption, showing that mayors who can run for reelection steal substantially fewer audited resources than lame-duck mayors and that the disciplining effect is strongest where voters' information and institutional punishment are otherwise weaker.

- The paper studies whether reelection incentives discipline politicians.
- The setting is Brazilian municipalities during 2001–2004.
- The authors construct direct corruption measures from CGU audit reports.
- First-term mayors can run again; second-term mayors face a term limit.
- First-term mayors are about 27 percent less corrupt in the preferred specification.
- Results are robust to alternative corruption measures, close-election RD, ability checks, and experience checks.

- The effect is stronger when local media and judiciary mechanisms are absent and when political competition is meaningful.
- Reelection incentives reduce visible corruption but do not reduce less visible mismanagement.
- First-term mayors also exert more effort to attract federal matching grants as elections approach.

10 Conclusion

10.1 Core conclusion

Ferraz and Finan show that electoral accountability reduces corruption. Mayors who can be punished by voters through the reelection channel misappropriate fewer public resources than mayors who face a binding term limit. The effect is economically meaningful and robust across measurement and identification strategies.

10.2 Economic intuition

The economic intuition is a political agency logic. A corrupt first-term mayor may prefer to behave as if he were honest because doing so increases the chance of reelection and access to future office rents. A lame-duck mayor does not face the same direct electoral incentive and therefore extracts more rents.

10.3 Incremental contribution revisited

The paper advances the corruption and political economy literature in three ways. First, it replaces perception-based corruption indexes with objective audit-based measures. Second, it identifies accountability effects within a single institutional setting rather than across countries. Third, it tests mechanisms by showing that the effect varies with information, judicial presence, political competition, and the visibility of violations.

10.4 Implications

The results imply that electoral rules, information provision, audits, and local accountability institutions can be complements. Elections discipline politicians only when voters have some capacity to observe or infer misconduct and when reelection is genuinely at stake. The findings also imply that anti-corruption policy should not focus only on punishment after corruption occurs; it should strengthen the incentives that make politicians avoid corruption in the first place.

10.5 Limitations

The paper estimates the effect of reelection incentives under Brazil’s two-term mayoral regime. It does not identify what would happen under unlimited reelection, three-term limits, or alternative electoral systems. The corruption measures are rich but still depend on what auditors detect and record. Finally, while the paper provides strong evidence on visible corruption, it is less informative about subtler forms of favoritism or policy distortion that audits may not capture.

10.6 Takeaway

The enduring takeaway is that political corruption is responsive to incentives. When elected officials expect future electoral consequences, they reduce visible rent extraction. Electoral accountability, especially when reinforced by information and institutional checks, can be a powerful constraint on political abuse.